

Organisms and their Environment

Question Paper 1

Level	IGCSE
Subject	Biology
Exam Board	CIE
Topic	Organisms and their Environment
Paper Type	(Extended) Theory Paper
Booklet	Question Paper 1

Time Allowed: 57 minutes

Score: /47

Percentage: /100

- 1 Moose, *Alces americanus*, shown in Fig. 6.1, are large herbivores that primarily live in northern parts of North America. They have a varied diet that includes young shoots of willow trees and aquatic plants.



Fig. 6.1

Isle Royale is a large island in Lake Superior in the United States where there is a population of moose that has been studied by ecologists for a long time. The animals' only predator is the wolf. The island has a population of wolves that has changed in numbers over the years.

- (a) (i) Draw a food chain for the organisms in the passages above.

[2]

- (ii) Complete Table 6.1 by stating the name and identifying the trophic level of each organism in the food chain.

Table 6.1

name of organism	trophic level

[3]

- (iii) State **two** factors that influence the numbers of a top predator, such as wolves.

1

2

[2]

- His results are summarised in Table 6.2.

energy input or output or energy flow	energy/MJ
consumed by moose	4 320 000
respiration of moose	380 000
consumed by wolves	56 000
respiration of wolves	53 000

- % [2]

- [5]

[Total: 14]

2 The human population is growing rapidly. This is increasing the pressure on the world food supply.

(a) Name the rapid growth phase of any population of organisms.

..... [1]

In Canada farmers are breeding fish in large nets because the wild stocks of fish are decreasing. Fig. 6.1 is a diagram of a salmon fish farm in the ocean. The salmon only eat the food provided by the worker.

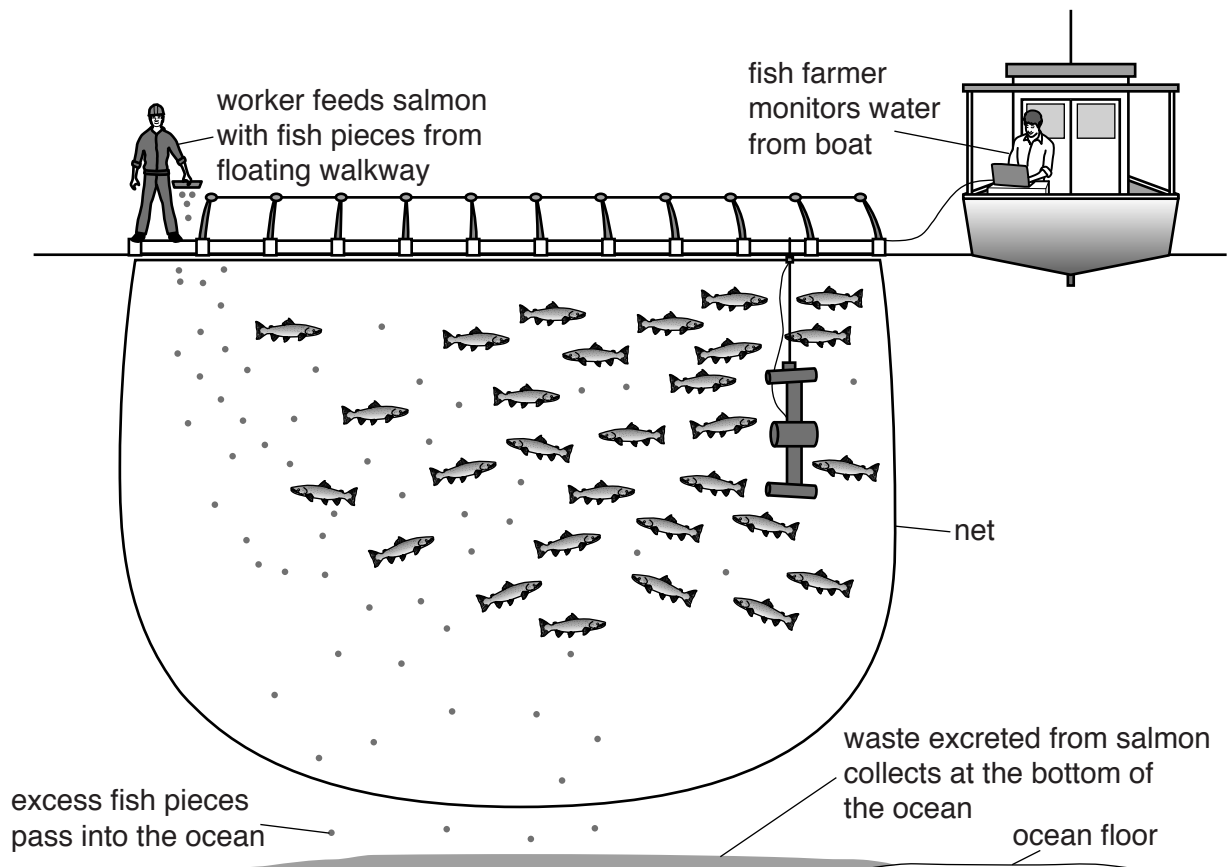


Fig. 6.1

- (b)** Explain the effects of the excess fish pieces and waste excreted from the salmon on the environment.

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..... [3]

- (c)** State the lowest possible trophic level of the salmon shown in Fig. 6.1.

..... [1]

- (d)** Other farmers grow seaweed. Salmon farming is a less energy efficient way of producing food for humans than seaweed farming.

Explain why.

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..... [3]

[Total: 8]

- 3 The pea plant, *Pisum sativum*, is a legume which is grown both as a human food and as livestock feed.

Fig. 6.1 shows some of the root nodules on a pea plant.

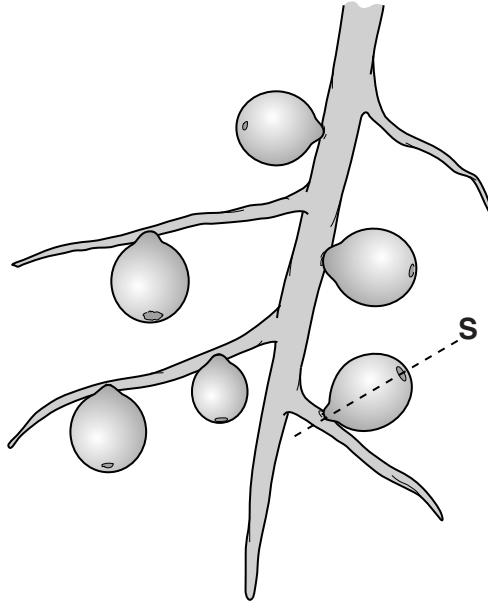


Fig. 6.1

Fig. 6.2 shows a cross-section through the root nodule at **S** on Fig. 6.1.

T indicates the transport tissue in the root.

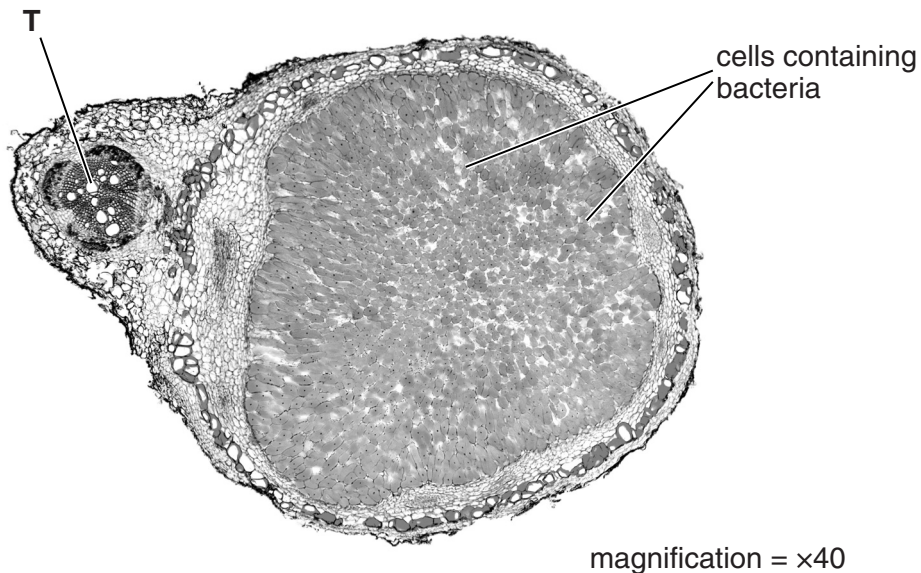


Fig. 6.2

- (a) The maximum diameter of the root nodule in Fig. 6.2 is 73 mm.

Calculate the actual diameter of the root nodule.

actual diameter [1]

- (b) Describe the role of the bacteria in the root nodules of *P. sativum*.

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.....
.....[2]

- (c) The bacteria require carbohydrates that are supplied by the pea plant.

Describe how the carbohydrates are produced and transported by the plant to the bacteria.

- (i) produced

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.....
.....[2]

- (ii) transported

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.....
.....
.....[2]

- (d)** Pea plants grow well in soils that are deficient in nitrogen.

Explain how root hair cells of pea plants absorb nitrate ions from soils with low nitrate concentrations.

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.....[2]

[Total: 9]

- 4 Fig. 1.1 shows the change in the biomass of the fungus *Penicillium* when grown in a fermenter to produce the antibiotic penicillin.

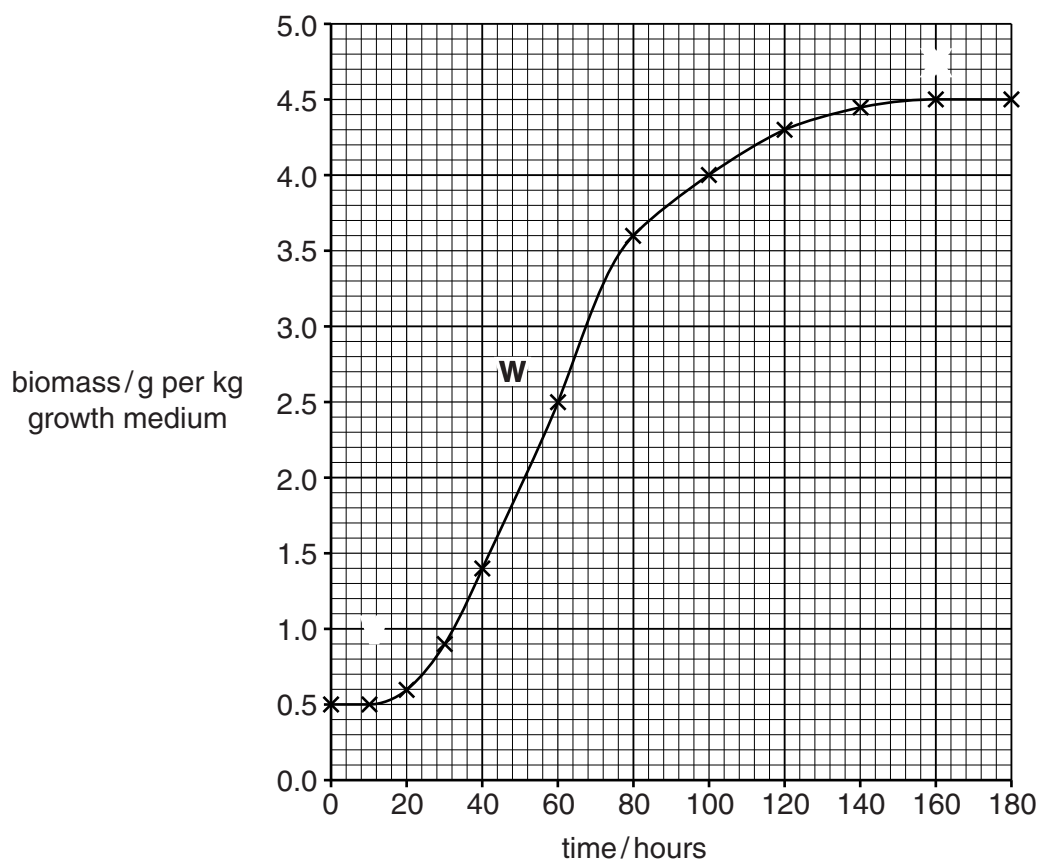


Fig. 1.1

- (a) Name the stages in the growth of *Penicillium* indicated by V, W and X.

V

W

X [3]

- (b) State **two** factors that are kept constant when *Penicillium* is grown in a fermenter.

1

2 [2]

- (c) Suggest why the growth of *Penicillium* is measured in biomass rather than numbers of cells.

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..... [1]

[Total: 6]